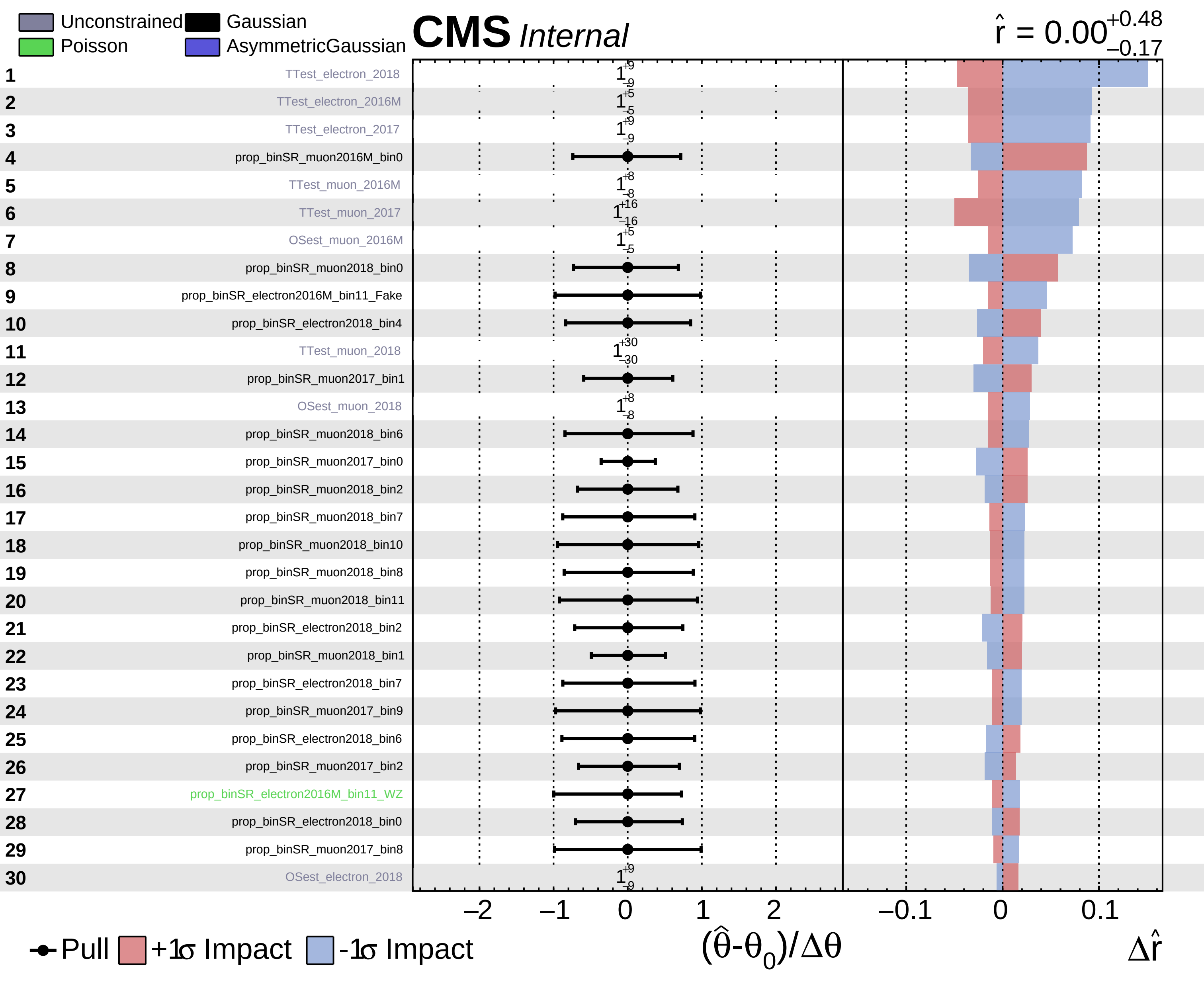
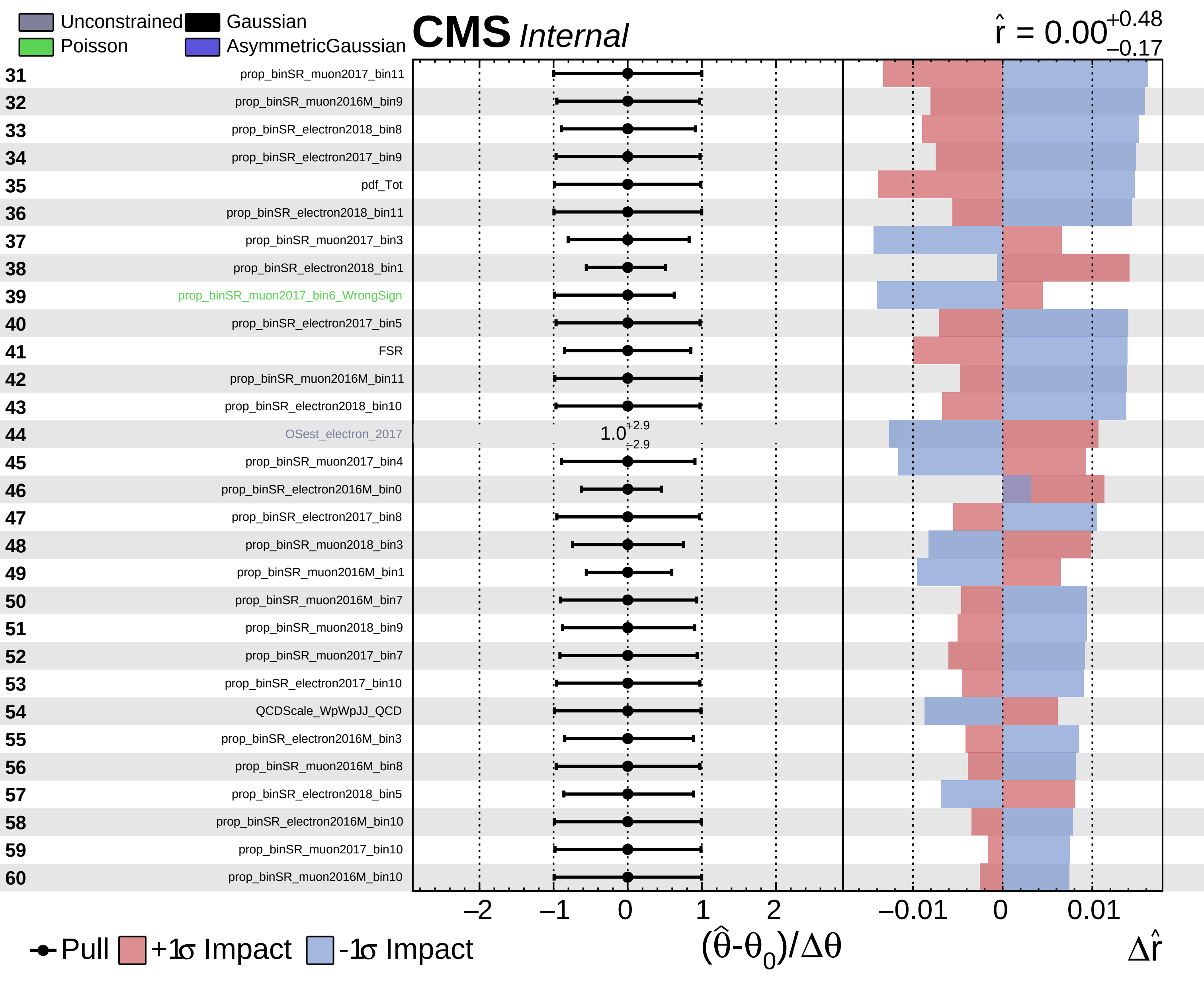
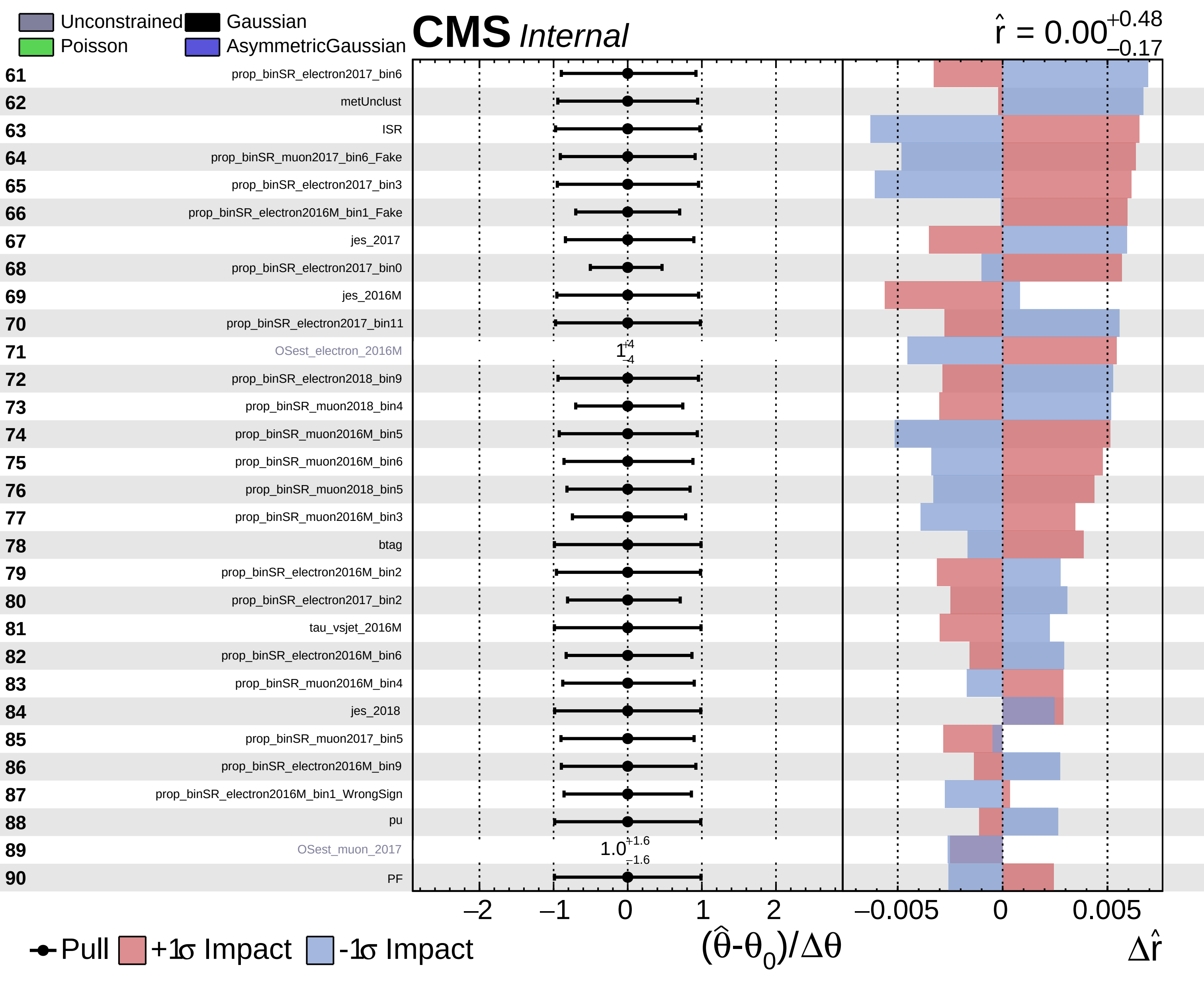


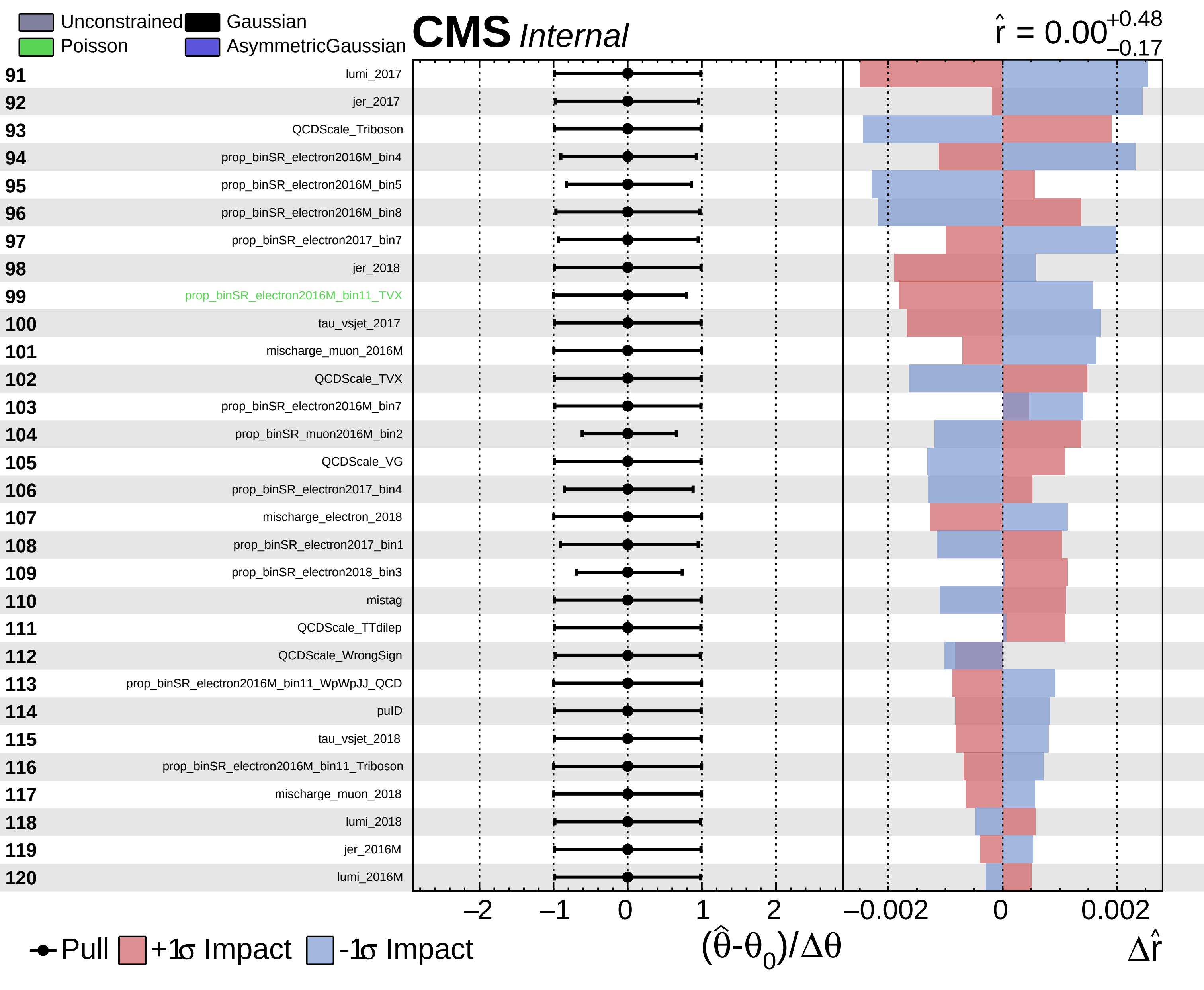
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 0.00^{+0.48}_{-0.17}$
1 TTest_electron_2018
2 TTest_electron_2016M
3 TTest_electron_2017
4 prop_binSR_muon2016M_bin0
5 TTest_muon_2016M
6 TTest_muon_2017
7 OSest_muon_2016M
8 prop_binSR_muon2018_bin0
9 prop_binSR_electron2016M_bin11_Fake
10 prop_binSR_electron2018_bin4
11 TTest_muon_2018
12 prop_binSR_muon2017_bin1
13 OSest_muon_2018
14 prop_binSR_muon2018_bin6
15 prop_binSR_muon2017_bin0
16 prop_binSR_muon2018_bin2
17 prop_binSR_muon2018_bin7
18 prop_binSR_muon2018_bin10
19 prop_binSR_muon2018_bin8
20 prop_binSR_muon2018_bin11
21 prop_binSR_electron2018_bin2
22 prop_binSR_muon2018_bin1
23 prop_binSR_electron2018_bin7
24 prop_binSR_muon2017_bin9
25 prop_binSR_electron2018_bin6
26 prop_binSR_muon2017_bin2
27 prop_binSR_electron2016M_bin11_WZ
28 prop_binSR_electron2018_bin0
29 prop_binSR_muon2017_bin8
30 OSest_electron_2018
1^{+9}_{-9}
1^{+5}_{-5}
1^{+9}_{-9}
1^{+8}_{-8}
1^{+16}_{-16}
1^{+5}_{-5}
1^{+30}_{-30}
1^{+8}_{-8}
1^{+9}_{-9}
−2 −1 0 1 2
−0.1 0 0.1
Pull +1σ Impact -1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$



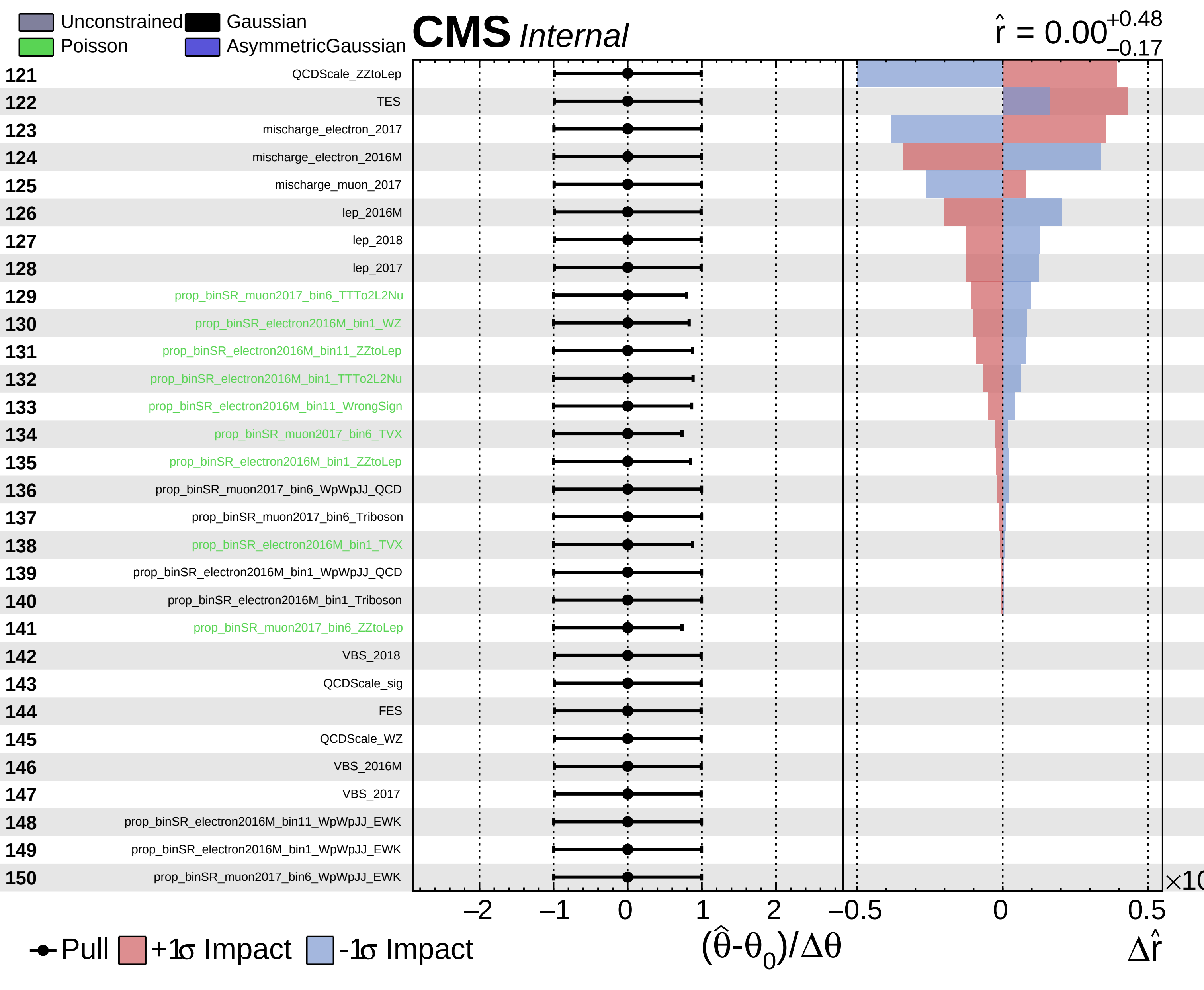
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
r̂ = 0.00 +0.48 −0.17
31 prop_binSR_muon2017_bin11
32 prop_binSR_muon2016M_bin9
33 prop_binSR_electron2018_bin8
34 prop_binSR_electron2017_bin9
35 pdf_Tot
36 prop_binSR_electron2018_bin11
37 prop_binSR_muon2017_bin3
38 prop_binSR_electron2018_bin1
39 prop_binSR_muon2017_bin6_WrongSign
40 prop_binSR_electron2017_bin5
41 FSR
42 prop_binSR_muon2016M_bin11
43 prop_binSR_electron2018_bin10
44 OSest_electron_2017
1.0 +2.9 -2.9
45 prop_binSR_muon2017_bin4
46 prop_binSR_electron2016M_bin0
47 prop_binSR_electron2017_bin8
48 prop_binSR_muon2018_bin3
49 prop_binSR_muon2016M_bin1
50 prop_binSR_muon2016M_bin7
51 prop_binSR_muon2018_bin9
52 prop_binSR_muon2017_bin7
53 prop_binSR_electron2017_bin10
54 QCDScale_WpWpJJ_QCD
55 prop_binSR_electron2016M_bin3
56 prop_binSR_muon2016M_bin8
57 prop_binSR_electron2018_bin5
58 prop_binSR_electron2016M_bin10
59 prop_binSR_muon2017_bin10
60 prop_binSR_muon2016M_bin10
−2 −1 0 1 2
−0.01 0 0.01
Pull
+1σ Impact
-1σ Impact
(θ̂-θ0)/Δθ
Δr̂



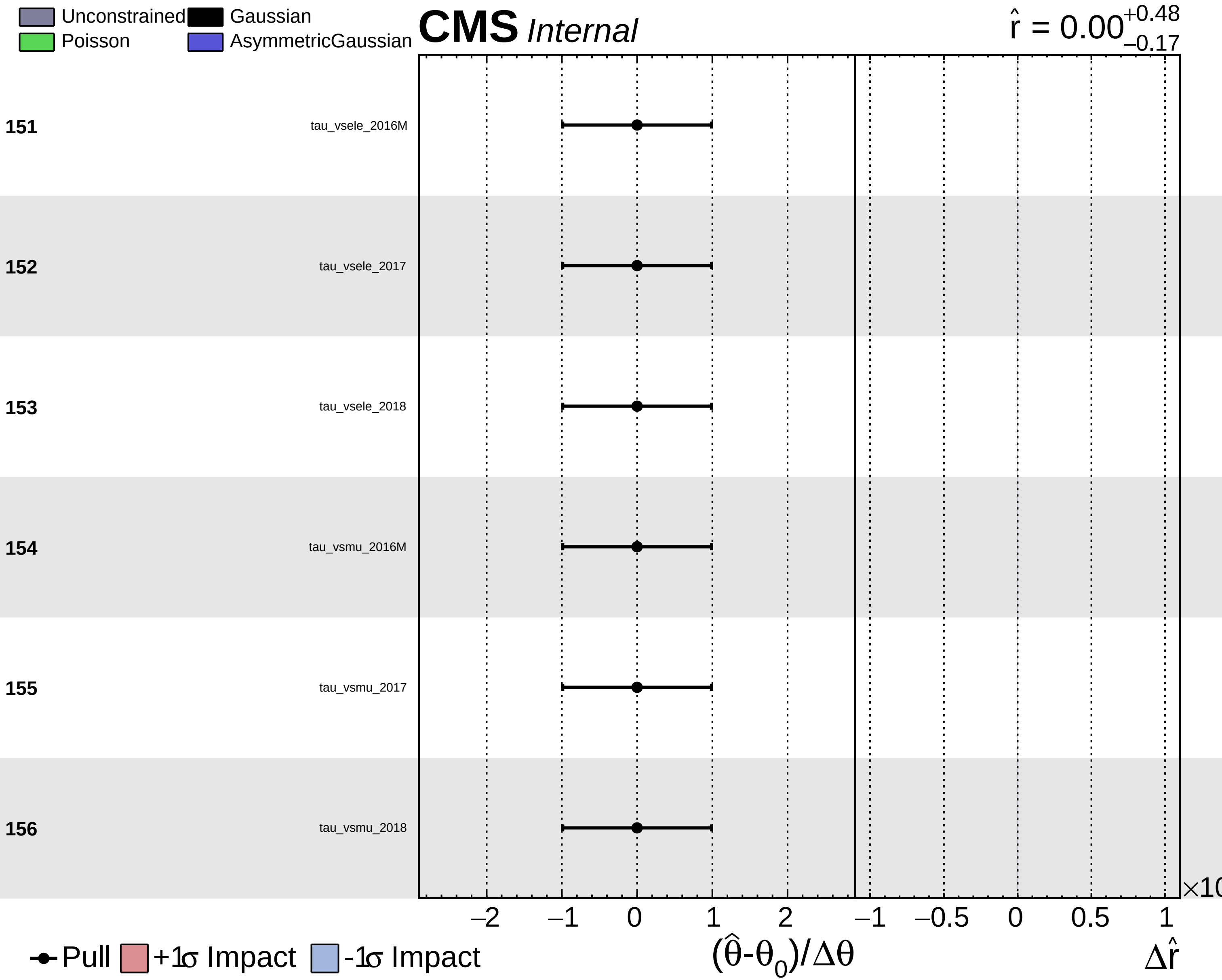
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 0.00^{+0.48}_{-0.17}$
61 prop_binSR_electron2017_bin6
62 metUnclust
63 ISR
64 prop_binSR_muon2017_bin6_Fake
65 prop_binSR_electron2017_bin3
66 prop_binSR_electron2016M_bin1_Fake
67 jes_2017
68 prop_binSR_electron2017_bin0
69 jes_2016M
70 prop_binSR_electron2017_bin11
71 OSest_electron_2016M
1^{+4}_{-4}
72 prop_binSR_electron2018_bin9
73 prop_binSR_muon2018_bin4
74 prop_binSR_muon2016M_bin5
75 prop_binSR_muon2016M_bin6
76 prop_binSR_muon2018_bin5
77 prop_binSR_muon2016M_bin3
78 btag
79 prop_binSR_electron2016M_bin2
80 prop_binSR_electron2017_bin2
81 tau_vsjet_2016M
82 prop_binSR_electron2016M_bin6
83 prop_binSR_muon2016M_bin4
84 jes_2018
85 prop_binSR_muon2017_bin5
86 prop_binSR_electron2016M_bin9
87 prop_binSR_electron2016M_bin1_WrongSign
88 pu
89 OSest_muon_2017
$1.0^{+1.6}_{-1.6}$
90 PF
−2 −1 0 1 2
−0.005 0 0.005
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$



Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 0.00^{+0.48}_{-0.17}$
91 lumi_2017
92 jer_2017
93 QCDScale_Triboson
94 prop_binSR_electron2016M_bin4
95 prop_binSR_electron2016M_bin5
96 prop_binSR_electron2016M_bin8
97 prop_binSR_electron2017_bin7
98 jer_2018
99 prop_binSR_electron2016M_bin11_TVX
100 tau_vsjet_2017
101 mischarge_muon_2016M
102 QCDScale_TVX
103 prop_binSR_electron2016M_bin7
104 prop_binSR_muon2016M_bin2
105 QCDScale_VG
106 prop_binSR_electron2017_bin4
107 mischarge_electron_2018
108 prop_binSR_electron2017_bin1
109 prop_binSR_electron2018_bin3
110 mistag
111 QCDScale_TTdilep
112 QCDScale_WrongSign
113 prop_binSR_electron2016M_bin11_WpWpJJ_QCD
114 puID
115 tau_vsjet_2018
116 prop_binSR_electron2016M_bin11_Triboson
117 mischarge_muon_2018
118 lumi_2018
119 jer_2016M
120 lumi_2016M
−2 −1 0 1 2
−0.002 0 0.002
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$



Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
r̂ = 0.00 +0.48 −0.17
121 QCDScale_ZZtoLep
122 TES
123 mischarge_electron_2017
124 mischarge_electron_2016M
125 mischarge_muon_2017
126 lep_2016M
127 lep_2018
128 lep_2017
129 prop_binSR_muon2017_bin6_TTTo2L2Nu
130 prop_binSR_electron2016M_bin1_WZ
131 prop_binSR_electron2016M_bin11_ZZtoLep
132 prop_binSR_electron2016M_bin1_TTTo2L2Nu
133 prop_binSR_electron2016M_bin11_WrongSign
134 prop_binSR_muon2017_bin6_TVX
135 prop_binSR_electron2016M_bin1_ZZtoLep
136 prop_binSR_muon2017_bin6_WpWpJJ_QCD
137 prop_binSR_muon2017_bin6_Triboson
138 prop_binSR_electron2016M_bin1_TVX
139 prop_binSR_electron2016M_bin1_WpWpJJ_QCD
140 prop_binSR_electron2016M_bin1_Triboson
141 prop_binSR_muon2017_bin6_ZZtoLep
142 VBS_2018
143 QCDScale_sig
144 FES
145 QCDScale_WZ
146 VBS_2016M
147 VBS_2017
148 prop_binSR_electron2016M_bin11_WpWpJJ_EWK
149 prop_binSR_electron2016M_bin1_WpWpJJ_EWK
150 prop_binSR_muon2017_bin6_WpWpJJ_EWK
−2 −1 0 1 2
−0.5 0 0.5
×10
Pull
+1σ Impact
-1σ Impact
(θ̂-θ0)/Δθ
Δr̂


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 0.00^{+0.48}_{-0.17}$
151
tau_vsele_2016M
152
tau_vsele_2017
153
tau_vsele_2018
154
tau_vsmu_2016M
155
tau_vsmu_2017
156
tau_vsmu_2018
−2
−1
0
1
2
−1
−0.5
0
0.5
1
×10
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$